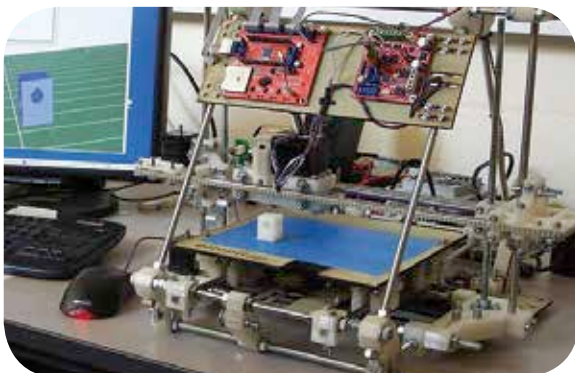


give rise to a whole generation of open-source hardware devices, where hobbyists or small companies could open source their hardware designs for mobile phones, tablets or laptops that can be further customized by end users and 'printed out'.

A cause of worry, however, is that such openly accessible manufacturing could pose security threats. Defense Distributed, an American company is in the business of developing and selling handheld gun designs that can be printed by users to manufacture their own working guns. Existing laws will need to be modified to ensure that new technologies like 3D printing don't end up causing more harm than good. In spite of some security concerns however, 3D printing, which is currently in its nascent stage, promises to provide the next breakthrough in the way the technology influences our lives.

Big Data

Big Data, as the name suggests quite tersely, is a collection of data sets so vast in size that it cannot be processed by traditional data processing and database management systems. This happens usually when the processing of this data, either for the purposes of spotting identifiable patterns (a process called data mining) or for searching, sharing or analysis is unable to keep up with the rate at which the raw data is being generated. Gartner defined Big Data as being governed by the "3V" framework - Volume, Velocity and Variety. This poses a unique problem with regards to the storage of such data, finding optimal techniques to parallelize the operations on



SMRC's prototype of a food-printer

this data and making useful interpretations based on such an analysis.

While Big Data is a term that is thrown around in a lot of fields including meteorology and geology, genomics and biological research, finance and business informatics, the one that interests us the most is the presence of such data in the internet space. Thanks to the ubiquity of internet connected, 'always-on' devices, internet companies like Google and Facebook are able to record and capture massive amounts of anonymous data regarding usage patterns. By analysing the behavioural patterns of internet users, Internet companies are able to create a more personalized experience for users, whether it be through targeted advertising, recommendations for new content, friend suggestions etc.

This form of 'learning' where a software makes recommendations based on your behaviour and choices is called 'Machine Learning', and the accuracy of these suggestions will only increase as this dataset is increased. The field of study is part of a larger field called 'Artificial Neural Networks', where an algorithm is designed to improve prediction accuracy with time as it identifies patterns in the user's behaviour. These are the algorithms that are at play when Google Now recommends a new restaurant or photospot, Last.fm recom-

mends a new friend or Grooveshark helps you discover new music based on your current playlists.

To deal with the technical difficulties of processing such large datasets, specialized systems have been invented, that attempt to reduce the time taken for these computations by parallelizing the computation tasks, and offloading the processing to hundreds or even thousands of servers. Apache Hadoop is one such open-source framework that aids in distributed computing by implementing Google's famous 'Map Reduce' algorithm, an algorithm that divides the work at hand into smaller fragments of work, each of which can be carried out by one node in the computing cluster, thereby reducing the total time required for computation.

As sensors and recording devices get more accurate, the volume of data being recorded is only going to keep increasing. Tackling this large influx of raw data is one of the primary problems for computer engineers, and scaling existing systems to cope with higher volumes of data is an area of intensive research. On the web, where a user's every move can, and is tracked, ideas and notions of privacy and anonymity are fast changing. Users will find they need to take extra care of their online actions, as internet accountability is gaining in importance. The applications of Big Data in health-care, genomics, geology and advertising holds a great deal of promise, and this is a field of computer science we'd all like to watch out for.. 



Putting things in perspective

1. Eric Schmidt claims that the volume of data generated every two days is equal to the data generated from the dawn of civilization to 2003. (source - <http://dgit.in/16UUeFh>)
2. If you piled a stack of CD-ROMs containing the world's current storage capacity, the stack would stretch about 80,000km beyond the moon. (source - <http://dgit.in/135bxy6>)

3. There are nearly as many bits of data in the world as there are stars in the entire universe (source - <http://dgit.in/11gF7gW>)
4. Each hour, the amount of information consumed online is enough to fill 7 million DVDs. (source - <http://dgit.in/10SJ8In>)
5. Of the nearly 250 billion emails that are sent everyday, almost 80% are spam. (source - <http://dgit.in/11gFc4b>)